
ALZHEIMER’S DISEASE CLASSIFICATION

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ABSTRACT

This study presents a deep learning pipeline for the classification of Alzheimer’s Disease (AD), Mild Cognitive Impairment (MCI), and Cognitively Normal (CN) patients using multi-modal MRI (T1, T2, FLAIR). By employing a robust preprocessing and early fusion strategy, we evaluated a custom 3D CNN, a focal-loss variant, and a MedicalNet 3D ResNet-18 transfer learning approach. Our baseline 3D CNN achieved the highest overall multi-class AUC (0.73), while the transfer learning model peaked at an AUC of 0.91 for the binary CN vs. AD task. Finally, an interpretability analysis with Grad-CAM confirmed the models’ reliance on clinically validated biomarkers.

1 Introduction

Alzheimer’s Disease (AD) is a progressive neurodegenerative disorder that causes brain atrophy and memory decline. Mild Cognitive Impairment (MCI) represents a transition stage between the healthy stage, also known as Cognitively Normal (CN) and AD. Diagnosing these stages from structural imaging is highly challenging due to subtle anatomical differences and high inter-subject variability.

The objective of this project is to develop a deep learning pipeline capable of classifying patients into CN, MCI, or AD using multi-modal Magnetic Resonance Imaging (MRI). In order to capture every relevant detail for the task, for each subject we utilize three distinct MRI sequences: T1-weighted (for structural and tissue details), T2-weighted (for fluid differences) and FLAIR (for white matter alterations and lesions).

In this study, we implement and compare multiple 3D Convolutional Neural Network (CNN) architectures. We begin with a custom lightweight 3D CNN baseline, address severe dataset class imbalances using weighted and Focal Loss, and ultimately leverage a Transfer Learning approach using the MedicalNet pre-trained backbone [1]. Finally, we perform an explainability analysis using Grad-CAM in order to verify that the models rely on clinically relevant biomarkers rather than background noise.¹

2 Related Works

In recent years, the classification of Alzheimer’s Disease has shifted from manual biomarker analysis to automated deep learning. Early approaches applied 2D Convolutional Neural Networks (CNNs) to individual MRI slices, but these methods often failed to capture the continuous 3D structure of brain atrophy [2].

In order to overcome this problem, researchers introduced 3D CNNs capable of processing the entire brain volume at once. Studies have shown that architectures like 3D-ResNets perform exceptionally well on structural MRI tasks by extracting deep volumetric features [3]. However, because neuroimaging datasets are typically small, training large 3D models from scratch often leads to severe overfitting. Most recently, studies have begun exploring 3D Vision Transformers (ViTs) and self-attention mechanisms to capture long-range anatomical dependencies across the brain

¹To ensure full reproducibility, the complete codebase, preprocessing scripts, and experimental notebooks are publicly available on GitHub at: <https://github.com/LeoMastro8/alzheimers-mri-classification>

[4, 5]. However, because Transformer architectures lack the inherent spatial inductive biases of CNNs, they are notoriously data-hungry and prone to severe overfitting on small clinical datasets.

To bypass the massive data requirements of Transformers while still preventing overfitting this project relies on Transfer Learning. By utilizing backbones pre-trained on large-scale medical datasets models, we can leverage prior anatomical knowledge rather than starting from random weights. This project directly builds upon these findings by comparing a custom 3D CNN against a pre-trained architecture.

3 Dataset and Input Preprocessing

The dataset is derived from ADNI-4 and contains a total of 952 patient sessions, provided in predefined training (664), validation (141), and test (147) splits. Each session includes multi-modal MRI scans (T1, T2, FLAIR). An initial inspection revealed a heavy class imbalance across all splits: approximately 50% of patients are CN, 40% are MCI, and only 10% are AD. This extreme minority of AD cases necessitated specific loss-weighting strategies during model training.

Because 3D neuroimaging data is highly complex and varies across different MRI scanners, a robust preprocessing pipeline was engineered to standardize the volumes and remove non-brain tissue. The preprocessing was executed on a Kaggle environment utilizing a dual NVIDIA T4 GPU setup, taking approximately 2 minutes per patient session. The sequential steps of our preprocessing are:

- **Skull Stripping (HD-BET):** The High-Dose Brain Extraction Tool (HD-BET) [6] is applied to the T1 volume. This neural network generates a highly accurate brain mask, stripping away irrelevant structures like the skull, eyes, and fat that could confuse the classification model, or make it learn irrelevant features.
- **Co-registration (ANTsPy):** To ensure perfect spatial alignment across all modalities, the T2 and FLAIR scans are rigidly co-registered to the T1 space using ANTsPy.
- **Standardization & Resizing (MONAI):** All volumes are reoriented to a standard canonical coordinate system using Nibabel, and then resized to a uniform dimension of $128 \times 128 \times 128$ voxels using trilinear interpolation via the MONAI framework.
- **Mask Application:** The resized T1 brain mask is multiplied against the resized T2 and FLAIR volumes.
- **Intensity Normalization:** A percentile based MinMax scaling is applied to map all voxel intensity values strictly into a $[0, 1]$ range, ensuring mathematical stability during training.

4 Model Architecture and Training Strategy

4.1 Multi-Modal Early Fusion Approach

To take advantage of the combined biological information of all the three MRI sequences, we utilized an Early Fusion strategy. During the dataset loading phase, the preprocessed T1, T2, and FLAIR volumes were stacked along the channel dimension. This created a unified 4D input tensor of shape $3 \times 128 \times 128 \times 128$ for each patient, allowing the models to learn cross-modal spatial correlations from the very first convolutional layer.

4.2 Baseline Architecture: Custom 3D CNN

The initial baseline model (EarlyFusion3DCNN) was a lightweight custom 3D Convolutional Neural Network consisting of roughly 1.16 million trainable parameters. The architecture features four sequential 3D Convolutional Blocks. Each block performs a $3 \times 3 \times 3$ convolution (padding of 1), followed by 3D Batch Normalization to stabilize training, a ReLU activation, and a $2 \times 2 \times 2$ Max Pooling layer to progressively downsample the spatial dimensions.

The channel depth increases through the blocks ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256$). After the final block, an AdaptiveAvgPool3d calculates the global average of each feature map layer reducing the spatial grid to a $1 \times 1 \times 1$ volume, leaving exactly 256 features to be fed into the classification head. The network concludes with a fully connected classification head with a Dropout layer ($p = 0.5$) for regularization, which outputs the final logits for the 3 classes.

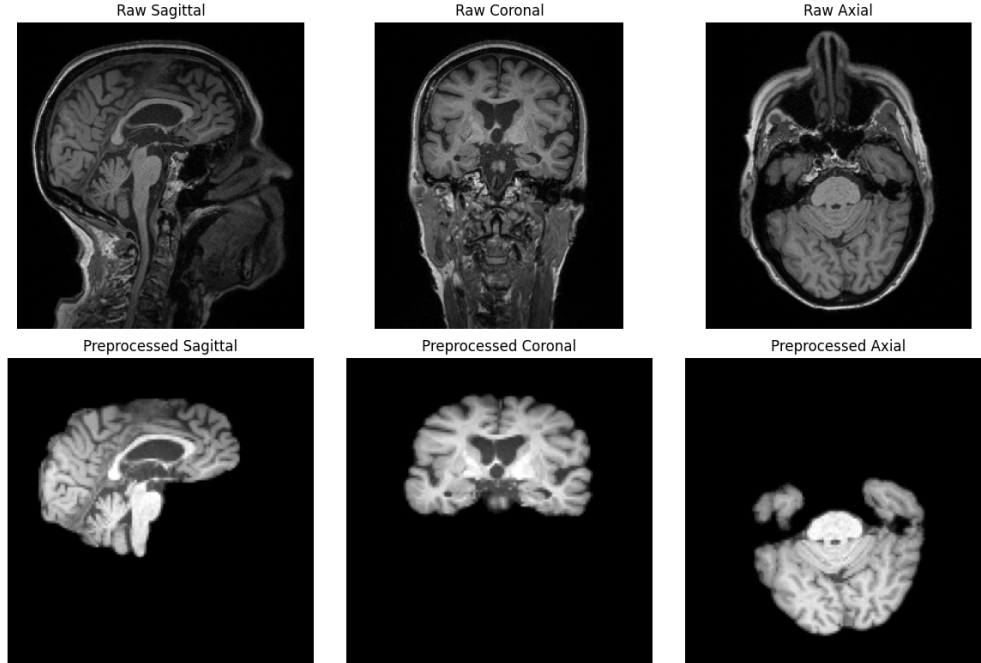


Figure 1: Visual output of the preprocessing pipeline: A T1-weighted MRI scan after skull stripping, canonical alignment, resizing to $128 \times 128 \times 128$ and intensity normalization.

4.3 Advanced Architecture: Transfer Learning with MedicalNet

To overcome the limitations of training a model from scratch on a small dataset, we leveraged a Transfer Learning approach by using the MedicalNet 3D ResNet-18 backbone [1]. This architecture was pre-trained on 23 different medical datasets, providing strong prior knowledge of 3D anatomical structures.

In order to adapt the model to our specific task:

1. The first convolutional layer (conv1) was modified to accept 3 input channels instead of 1. The pre-trained single-channel weights were divided by 3 and copied across the three new channels. This ensures the model retains its pre-trained knowledge rather than starting from random noise.
2. The final classification head was replaced with a randomly initialized linear layer mapping to our 3 target classes (CN, MCI, AD).

4.4 Training Strategy and Overfitting Control

Medical datasets are prone to severe overfitting due to high feature dimensionality and small sample sizes. We implemented strict regularization and specialized loss functions across our experiments:

Data Augmentation: To simulate scanner variance and prevent the model from memorizing specific brains, we applied subtle 3D augmentations during training using MONAI. These included random affine transformations (maximum 5° rotation and 5-voxel translation to avoid edge clipping), random intensity scaling (up to 10%), and Gaussian noise (up to 10%).

Handling Class Imbalance: The severe lack of AD patients might cause baseline models to ignore the minority class. To mitigate this, we implemented two mathematical strategies:

- **Inverse Frequency Weighting:** Applied inside the standard Cross-Entropy loss to heavily penalize errors on the minority class. The weight for each class c was computed using the formula $W_c = \frac{N_{total}}{N_{classes} \times N_c}$. This resulted in a balancing tensor of [0.6453, 0.8137, 4.5170] for CN, MCI, and AD, respectively, forcing the network to treat a single AD misclassification as roughly seven times more severe than a CN error.
- **Focal Loss** ($\gamma = 2.0$): Evaluated in our second experiment, this dynamically scales the weighted Cross-Entropy loss based on prediction confidence using the formula $FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$. By modulating the

standard loss with the $(1 - p_t)^\gamma$ factor, the optimizer down-weights easy-to-classify examples and focuses its gradients almost entirely on "hard" boundary cases.

We conducted three different training runs to evaluate different loss functions and optimizers. All model training was executed in a Kaggle environment utilizing a dual NVIDIA T4 GPU setup with an effective batch size of 8. In terms of computational cost, the custom 3D CNN models required approximately 1 hour to reach convergence. In contrast, the MedicalNet Transfer Learning model required roughly 2 hours due to the increased architectural depth. Across all runs, an Early Stopping approach against the Validation Balanced Accuracy was used to halt training immediately when generalization ceased. The hyperparameters for all the runs are summarized in Table 1.

Table 1: Hyperparameter Configurations Across Training Runs

Parameter	Baseline (3D CNN)	Focal (3D CNN)	MedicalNet (ResNet-18)
Batch Size	8	8	8
Optimizer	Adam	Adam	Adam
Learning Rate (LR)	1×10^{-4}	2×10^{-4}	5×10^{-6}
Weight Decay (L2)	1×10^{-4}	1×10^{-3}	1×10^{-4}
Loss Function	Weighted CE	Focal Loss ($\gamma = 2.0$)	Weighted CE
Early Stop Patience	15 Epochs	15 Epochs	7 Epochs

For the MedicalNet, we drastically lowered the learning rate to 5×10^{-6} . This change was critical to prevent a typical Transfer Learning phenomenon called "Catastrophic Forgetting", ensuring that large gradient updates did not overwrite the pre-trained weights in the ResNet backbone. All models utilized a ReduceLROnPlateau scheduler (factor of 0.5) to fine-tune the weights as validation accuracy plateaued.

5 Results and Explainability Analysis

5.1 Quantitative Results

The models were evaluated on an independent test set of 147 patient sessions. We evaluated multi-class (ternary) performance alongside binary tasks. Rather than retraining separate binary models from scratch, which is very computationally expensive and wastes the broader anatomical context the model already learned, we extracted binary metrics directly from the 3-class models using conditional probability. To evaluate a specific pair, such as CN vs. AD, we first filtered the test set to exclude any MCI patients. Next, we took the model's 3-class predictions for the remaining patients and ignored the MCI probability. We then re-normalized the probabilities of the two target classes so they sum to 1.0 (100%). This is expressed as: $P(AD | CN \cup AD) = \frac{P(AD)}{P(CN) + P(AD)}$. This method allowed us to accurately isolate the model's binary discriminative power without the massive computational overhead of retraining.

Table 2: Model Performance Metrics on the Test Set

Evaluation Metric	Baseline 3D CNN	Focal Loss ($\gamma = 2$)	MedicalNet (Transfer)
<i>Multi-Class (Ternary) Performance</i>			
Accuracy	0.5986	0.3401	0.4898
Balanced Accuracy	0.4583	0.5152	0.4471
Macro Precision	0.4705	0.5183	0.4454
Macro Recall	0.4583	0.5152	0.4471
Macro F1-Score	0.4575	0.3250	0.4440
Macro Specificity	0.7638	0.7283	0.7040
AUC (OVR)	0.7327	0.6883	0.6929
<i>Binary Task Performance (AUC)</i>			
CN vs. MCI	0.6554	0.6131	0.5926
MCI vs. AD	0.8770	0.8704	0.8770
CN vs. AD	0.8634	0.8449	0.9155

As shown in Table 2, the Baseline 3D CNN yielded the most robust ternary classification, achieving an overall Accuracy of 59.86% and an OVR AUC of 0.7327. The MedicalNet model struggled slightly with the ternary task but achieved the highest binary AUC of 0.9155 for distinguishing Healthy (CN) from diseased (AD) brains.

The application of Focal Loss aggressively penalized the model for missing the minority AD and ambiguous MCI classes. While this successfully raised the Balanced Accuracy to 51.52%, it sacrificed overall accuracy (dropping to 34.01%), highlighting the extreme difficulty the network faced in resolving the subtle anatomical transition between CN and MCI.

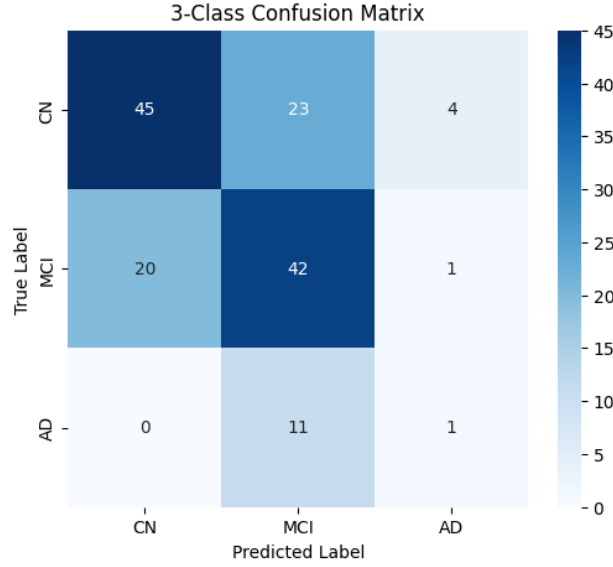


Figure 2: Confusion Matrix for the Baseline 3D CNN.

5.2 Explainability and Interpretability Analysis

Model interpretability is mandatory in medical imaging, and it is used to ensure clinical trustworthiness. In order to verify that our networks were relying on actual biological structures rather than background noise, we utilized Captum’s Layer Grad-CAM to visualize the spatial attention of the 3D convolutional layers.

The resulting Grad-CAM heatmaps demonstrated that the models successfully learned clinically validated biomarkers without any explicit spatial supervision. For example, when classifying Alzheimer’s Disease (AD) patients, the model’s attention localized intensely in the center of the brain, specifically targeting the lateral ventricles. This qualitative visual verification confirmed that our model makes its predictions based on sound medical realities, proving it is highly effective at extracting disease-specific features.

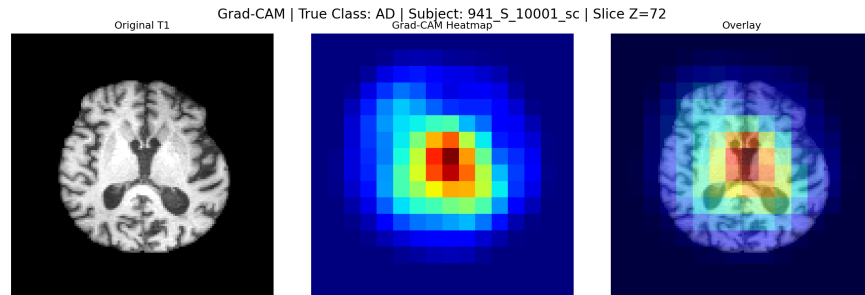


Figure 3: Grad-CAM overlay for an AD patient. The red regions highlight the model’s focus on the expanded lateral ventricles.

6 Conclusion

In this work, we built an end-to-end multi-modal 3D computer vision pipeline for Alzheimer’s Disease classification. By effectively preprocessing and fusing T1, T2, and FLAIR volumes, our models extracted deep volumetric features without relying on manual biomarker tracing. The custom Baseline 3D CNN proved highly effective (0.73 multi-class AUC), while transfer learning via MedicalNet demonstrated outstanding binary discriminative power (0.91 AUC for CN vs. AD). Finally, interpretability analysis confirmed that the networks independently learned to target clinically relevant markers.

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